

Pierre Chamoun

 [pierre-chamoun](#) |  [pierre-chamoun](#) |  [pierre-chamoun.github.io](#) |  pierre.chamoun@mail.mcgill.ca

EDUCATION

McGill University

Expected Dec 2027

Master of Science (M.Sc.), Robotics

GPA: 4.00/4.00 | FRQNT Master's Scholarship | McGill Engineering Undergraduate Student Master's Award (MEUSMA)

McGill University

Dec 2025

Bachelor of Engineering (B.Eng.), Mechanical Engineering

GPA: 3.97/4.00 | J.W. McConnell Scholarship | NSERC Research Award | Dean's Honour List | McGill Robotics

TECHNICAL SKILLS

Programming	C++, Python, MATLAB
Robotics & Tools	ROS, GTSAM, Docker, Git
Technical Areas	State Estimation, Control Theory, Computer Vision, Machine Learning
Other	Scientific Computing, Statistics, GD&T, CAD

WORK EXPERIENCE

DECAR Group, McGill University

Jan 2026 – Present

Graduate Researcher in Robotics

- Investigating targetless multi-sensor calibration in unstructured environments for autonomous mobile robots.

Pratt & Whitney Canada

May 2025 – Aug 2025

Engineering Internship in Thermal Analysis and CFD, Turbine Rotating Structures

- Created a CATIA model of a rotating heat transfer test rig originally used in the 1980s to develop one-dimensional correlations for heat transfer coefficients on rotating turbine discs.
- Developed a CFX computational fluid dynamics (CFD) model of rotating turbine discs and calibrated it to match heat transfer measurements obtained from thermocouples at specific locations.
- Derived three-dimensional heat transfer correlations from the CFD simulation results.

Pratt & Whitney Canada

Jan 2025 – Apr 2025

Engineering Internship in Mechanical Design

- Introduced engineering changes to reduce non-conformities resulting from the manufacturing process.
- Validated and prepared documentation for various engineering change requests.

McGill Aerodynamics Laboratory

May 2024 – Aug 2024

Research Internship in Computational Fluid Dynamics

- Applied machine learning techniques using Python to study a hydrodynamic field resulting from a simulation of a turbulent oscillatory internal flow.
- Trained a surrogate model to predict the temporal dynamics of the velocity field, reducing reliance on costly direct numerical simulations.

McGill Aerodynamics Laboratory

May 2023 – Aug 2023

Research Internship in Computational Fluid Dynamics

- Conducted direct numerical simulations on the Graham compute cluster using the C++ open-source software Channelflow to study turbulent internal flows subjected to an oscillatory pressure gradient.
- Performed data analysis with Python using statistical and spectral methods.